NIFTY Short Iron Condor Algo

Build plan for a Python paper-trading system on Zerodha Kite Connect, running on localhost

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What this plan covers

Your rules as a precise spec, the points in them that need a decision before coding, the daily and weekly run schedule, the system design (modules, data, fill model, exits, dashboard), a phase-by-phase build order, and the checks to pass before any real money goes in.

Strategy rules as I understand them

Rule	Spec
Instrument	NIFTY weekly options. Lot size 65; weekly expiry is Tuesday (both confirmed from current sources).
Structure	Short Iron Condor: sell 1 OTM CE + 1 OTM PE, buy 1 further-OTM CE + 1 further-OTM PE as hedges.
Pre-market filter	Check India VIX and NIFTY gainers/losers before the open. Trade only if VIX < 20.
Width	500 to 600 points, chosen by days to expiry (see the open questions on the next page).
Hedge length	200 to 300 points between each short strike and its hedge.
Stop loss	Exit everything when total P&L reaches −₹250 to −₹300.
Target	Exit everything when total P&L reaches +₹400 to +₹500.
Parallel tracking	Track the combined strategy premium and NIFTY live, like your straddle chart, and use premium movement as an extra exit signal.
Trade days	Expiry day plus the 2 trading days before it: normally Fri, Mon, Tue. Never the day after expiry (Wed). That leaves Thu as a no-trade day too.
Mode	Paper trading on live Kite data; no real orders in v1.

1. Decisions to make before coding

These are gaps or ambiguities in the rules. Each has a default I'd build with; the config file lets you change it later without touching code.

#	Question	Default I'd use
1	Does “width 500–600” mean the gap between the short CE and short PE (so each short is 250–300 pts from spot), or each short’s distance from spot?	Gap between the shorts, centred on ATM. Expiry day (0 DTE): 500. 1 DTE: 550. 2+ DTE: 600.
2	Which hedge length goes with which day?	Expiry day 200, 1 DTE 250, 2+ DTE 300. Wider hedges cost more but protect more when there is more time left.
3	Is the ₹ P&L per lot or for the whole position, and before or after charges?	For the whole position, after estimated charges. Charges matter a lot here (see the note below).
4	What do gainers/losers actually decide? Your rules say to check them but give no rule.	Log them in v1 and add a soft filter: skip if NIFTY gaps more than ±0.8% in pre-open or advance/decline is extremely one-sided (e.g. 45+ of 50 in one direction). Tune after 4–6 weeks of data.
5	What exactly is the premium-movement exit?	Exit if the combined net premium rises 35% above the entry credit, or if either short leg’s price doubles. This catches a fast trending move before the ₹ stop is hit through slippage.
6	Entry time and end-of-day handling.	Enter at 09:20 after the opening auction settles. If neither stop nor target is hit, square off at 15:15. No overnight carry in v1.
7	One trade per day, or re-enter after an exit?	One trade per day. Re-entry adds risk and should be tested separately later.

Important: the ₹ thresholds are very small for 1 lot

With 1 lot of 65, a +₹500 target is only about 7.7 points of combined premium decay, and a −₹300 stop is about 4.6 points. On 4 legs, the bid–ask spread alone can be 2–4 points, so the stop may trigger on noise within minutes of entry.

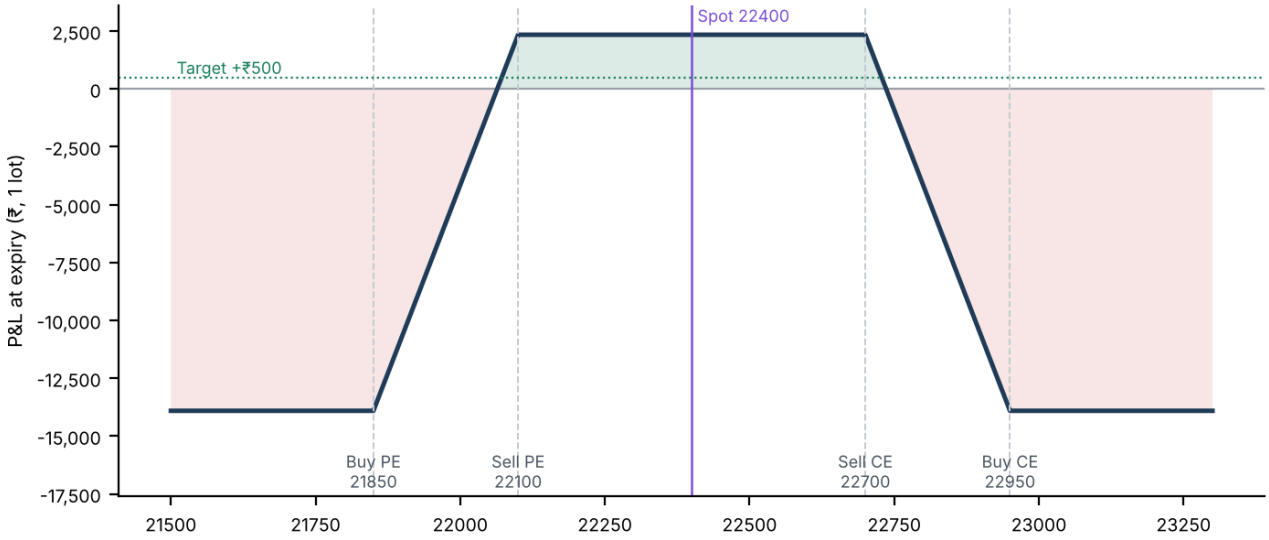
Charges also take a large share: 8 executed orders × ₹20 brokerage + GST is roughly ₹190, before STT and exchange fees. That means a +₹500 gross win nets close to ₹250–₹300.

Recommendation: define the stop and target in **points of net premium** (e.g. target 8 pts, stop 5 pts) and compute ₹ = points × 65 × lots. Paper trading will show quickly whether these levels are workable. The system will log both gross and net P&L so you can see the real effect.

2. A worked example

Using NIFTY at 22400 (from your screenshot), a 600-wide condor with 250-point hedges. Premiums are illustrative, not live quotes.

Leg	Strike	Action	Premium (illustrative)
Hedge put	21850 PE	Buy	12
Short put	22100 PE	Sell	30
Short call	22700 CE	Sell	28
Hedge call	22950 CE	Buy	10
Net credit			**36 pts = ₹2,340 per lot**



Max profit at expiry is ₹2,340; max loss is (250 − 36) × 65 ≈ ₹13,900 if NIFTY ends beyond either hedge. Your intraday stop at −₹300 is meant to keep you far from that, but gaps and fast moves can push the fill past the stop. That gap risk is the main reason the hedges must always be in place before the shorts.

3. When the system runs

Weekly calendar

Day	Status	Days to expiry	Width / hedge (default)
Wednesday	No trade (day after expiry)	6	–
Thursday	No trade (outside the 3-day window)	5	–
Friday	Trade	2	600 / 300
Monday	Trade	1	550 / 250
Tuesday (expiry)	Trade	0	500 / 200

Holidays move the expiry. The algo must not hard-code Tuesday: it reads the nearest NIFTY expiry from Kite's instruments list each morning, builds the list of trading days before it (skipping NSE holidays), and decides from that. If expiry moves to Monday, the window becomes Thu, Fri, Mon.

Daily timeline (IST)

Time	Step	What happens
08:00	Login	Kite access tokens expire daily, so you log in once each morning; a small local page captures the request token and saves the session.
08:15	Day check	Load instruments, find nearest expiry, decide whether today is a trade day. If not, log and stop.
09:00–09:08	Pre-open	Read India VIX, NIFTY pre-open price and the 50 constituents' quotes. Compute gap %, gainers/losers and advance/decline.
09:15	Open	Start the WebSocket feed for NIFTY, VIX and the candidate strikes.
09:20	Entry	Apply the VIX < 20 filter, pick strikes, paper-fill the hedges first and then the shorts.
09:20–15:15	Monitor	On every tick: recompute P&L and net premium, check exits in priority order, update the dashboard.
15:15	Time exit	Square off anything still open.
15:35	Report	Write the trade journal entry, daily summary and charts.

Getting the pre-market data

- **India VIX:** available from Kite as an index quote (`NSE:INDIA VIX`). No scraping needed.
- **Gainers/losers:** nseindia.com actively blocks scripted access and changes its endpoints, so it breaks often. More reliable: pull quotes for the 50 NIFTY constituents from Kite during pre-open and rank them yourself. NSE scraping can be an optional fallback.
- **Data plan:** live quotes, WebSocket ticks and historical candles need the paid Kite Connect plan (₹500/month per API key). The free Personal plan has no market data, so it can't run this system.

4. System design

Small, single-purpose modules, so the paper broker can later be swapped for a live broker without changing any strategy code.

Module	Responsibility
config.yaml	Every tunable number: VIX limit, widths, hedges, stop/target, premium exit %, times, lots, charge rates, paper/live flag.
auth.py	Daily Kite login flow; stores the access token locally (never commit it).
instruments.py	Downloads the NFO instrument dump daily; resolves expiry, strikes and tokens; holiday-aware trade-day logic.
premarket.py	VIX, gap %, gainers/losers, advance/decline; returns TRADE or SKIP with a reason.
market_data.py	KiteTicker WebSocket in full mode (best bid/ask needed for fills); keeps an in-memory latest-tick table; auto-reconnect.
strike_selector.py	Rounds spot to the nearest 50 for ATM, applies width and hedge, checks liquidity (spread, OI) and minimum credit.
broker/paper.py	Simulated fills at bid/ask plus slippage, charge model, position book. Same interface as the future broker/live.py.
risk_manager.py	Runs the exit rules in priority order each tick; kill switch; one trade per day; max daily loss.
strategy.py	Orchestrates the day: check → enter → monitor → exit → report.
storage.py	SQLite: trades, legs, fills, a tick snapshot every 5 s, daily summary.

Module	Responsibility
dashboard/	Local web page (FastAPI + a JS chart) showing live net premium vs NIFTY, P&L, legs and exit levels.
backtest/	Phase 4: replays historical candles through the same strategy and risk code.

Strike selection logic

```
atm      = round(spot / 50) * 50
width    = cfg.width_by_dte[dte]          # 600 / 550 / 500
hedge    = cfg.hedge_by_dte[dte]          # 300 / 250 / 200
short_pe = atm - width // 2               # rounded to a listed strike
short_ce = atm + width // 2
long_pe  = short_pe - hedge
long_ce  = short_ce + hedge

reject if any leg's (ask - bid) > cfg.max_spread_pts
reject if net_credit < cfg.min_credit_pts   # not worth the risk + charges
order: BUY long_pe, BUY long_ce, then SELL short_pe, SELL short_ce
```

Buying the hedges first matches how margin and risk work on the real exchange, so the paper results stay realistic.

Exit engine (checked every tick, first match wins)

Priority	Rule	Trigger (default)
1	Kill switch	Manual button on the dashboard, data feed stale for more than 10 s, or an unexpected error.
2	Stop loss	Net P&L ≤ −₹300 (or −5 pts × 65 × lots if you switch to points).
3	Premium blow-out	Net premium ≥ entry credit × 1.35, or any short leg ≥ 2× its entry price.
4	Target	Net P&L ≥ +₹500.
5	Time exit	15:15.

P&L is calculated as if every leg were closed now at the price you would actually get (shorts at the ask, hedges at the bid), not at LTP. This is more pessimistic, and it is what you would really receive.

Paper fill model

- Sell at the best bid, buy at the best ask, then apply an extra slippage of 0.5 point per leg (configurable).
- If the depth is empty or the spread is wider than the limit, the order is rejected and logged, the same way a real limit order would sit unfilled.
- Charges per order: ₹20 brokerage + 18% GST, STT on the sell-side premium, exchange transaction fee, SEBI fee and stamp duty. Keep these rates in config, because they change.

Live tracking view (like your chart)

- Top panel: combined net premium (short CE + short PE − long CE − long PE) vs NIFTY, on the same time axis, updated every second.
- Horizontal lines for the entry credit, the premium-blowout level, and the premium levels that correspond to the ₹ stop and target.
- Bottom panel: running P&L in ₹ (gross and net of charges), and a table of the 4 legs with entry, current bid/ask and P&L.
- A history view to replay any past day from the 5-second snapshots.

Project layout

```
iron_condor/
  config.yaml            .env  (API key, secret - git-ignored)
  main.py                # runs the day: python main.py
  core/   auth.py  instruments.py  premarket.py  market_data.py
          strike_selector.py  risk_manager.py  strategy.py  storage.py
  broker/ base.py  paper.py  (live.py later)
  dashboard/  app.py  static/index.html
  backtest/   engine.py
  tests/      test_strikes.py  test_exits.py  test_fills.py  test_calendar.py
  data/       trades.db  logs/
```

Libraries: kiteconnect, pandas, pyyaml, fastapi + uvicorn, APScheduler (timed steps), pytest. Python 3.11+.

5. Build phases

Phase	Deliverable	Done when	Time
0. Setup	Kite paid API key, project skeleton, config, daily login working.	You can fetch a NIFTY quote and India VIX from Python.	1–2 days
1. Data	Instruments + trade-day calendar, pre-market module, WebSocket feed with reconnect.	A full day of ticks for NIFTY, VIX and 20 strikes is saved without gaps.	3–4 days
2. Core logic	Strike selector, paper broker with charges, risk manager, strategy loop, SQLite storage.	Unit tests pass for strikes, every exit rule, fills and the calendar (including holiday weeks).	1 week
3. Dashboard	Live chart + legs table + kill switch + daily report.	You can watch a paper trade live and replay it afterwards.	3–4 days
4. Paper run	Run every trade day; weekly review of the journal.	At least 6–8 weeks (18–24 trade days) of clean logs.	6–8 weeks
5. Backtest	Replay historical data to test other widths, stops and VIX limits.	Parameter choices are backed by more than a few weeks of live paper results.	1–2 weeks (parallel with 4)

6. Before going live

- Paper results are positive **after charges**, and the worst day and worst losing streak are amounts you can accept.
- Paper fills compared with real fills on a few manual 1-lot trades, so you know how optimistic the paper model is.
- Live orders use limit orders, or market orders with market protection (Kite requires market protection on API market orders). Hedges are always filled before shorts, and if a hedge fails to fill the shorts are never sent.
- A reconciliation check every minute: positions in Kite must match the algo’s position book, or the kill switch fires.
- Margin check before entry using Kite’s basket margin API.
- Review SEBI’s current rules for retail algo trading through brokers (static IP, order-rate limits, algo registration) and Zerodha’s current API terms before placing automated live orders. These rules have been changing, so confirm them at the time.
- Start live with 1 lot, keep the same config as paper, and change only one parameter at a time.

> **Next step**
> Answer the 7 questions in section 1 (or accept the defaults). Then I'll build Phase 0–2: the project skeleton, config, login, data feed, strike selector, paper broker and exit engine, with tests.

This plan is for building and testing software. It is not financial advice. Short options carry large losses if markets gap, and past paper results don’t guarantee live results.